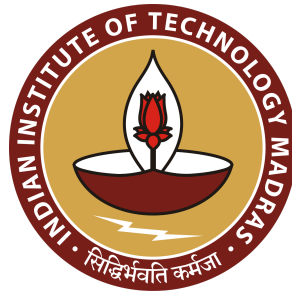


GRAND CHALLENGE - 2025

Integration of Verilog IP without AXI4Lite with Shakti SoC



DEVELOPED BY :
SHAKTI DEVELOPMENT TEAM @ IITM
SHAKTI.ORG.IN

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0.1 Proprietary Notice

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0.2 Release Information

Version	Date	Updates	Reviewed by
1.0	01/11/2025	Poovarasan M	Kotteeswaran E

Introduction

This document details the step by step procedure to integrate the **Custom Verilog IPs without AXI4Lite** interface with Shakti SoC. It is assumed that the verilog IP that needs to be ported does not have AXI4/AXI4Lite interface. The AXI4Lite slave interface needs to be added using this document.

The document uses GPIO verilog IP as an example to explain the overall flow. In this GPIO IP, the AXI4Lite Slave interface logic will be appended and added to Shakti IP before running bit stream generation.

This document gives the details of the verilog files used for this integration and the directory it is present.

To summarize, the GPIO Verilog IP without any bus protocol support is taken. The AXI4 Lite Slave logic is added to program the registers. Now the GPIO Verilog IP turns out to support the AXI4Lite Slave integration. After this, the **“Verilog_IP_Integration_with_Shakti_SoC”** document will guide to integrate the GPIO Verilog IP with AXI4Lite Slave interface with Shakti SoC.

Integration of GPIO Verilog IP

File: “gc2025/hw/ip_verilog/gpio.v”

This file is the GPIO Verilog IP without any bus protocol logic present.

File: “gc2025/hw/ip_verilog/gpio_tb.v”

Test bench to verify the functionality of GPIO IP using Simulation tool.

File: “gc2025/hw/ip_verilog/gpio_axilite.v”

AXI4Lite logic appended with the “gpio.v” file. The appended logic for AXI4Lite integration is the only thing needed to be picked up and used in your Custom IP. Write and Read logic of registers alone will be changed based on the register size present in the IP.

File: “gc2025/hw/ip_verilog/gpio_defs.v”

Verilog file to define the offsets for the registers used in the GPIO.